

Multi-Modal Agentic AI System for Medical Diagnosis and Triage

Project Timeline

AUGUSTINE CHIBUZOR AKAUBA

STUDENT ID: 24127523

<https://github.com/abhishek7112000/Multi-Modal-Agentic-AI-System-for-Medical-Diagnosis-and-Triage>

1. Project Overview

This document presents the full project timeline for the Multi-Modal Agentic AI System for Medical Diagnosis and Triage. The system autonomously diagnoses chest and cardiovascular conditions by combining three machine learning models, a medical large language model (MedGemma-27B-IT), and a retrieval-augmented generation (RAG) layer. The pipeline processes chest X-ray images, structured electronic health records (EHR), and free-text clinical notes, and delivers a triage decision with specialist referral recommendation.

2. Architecture Summary

The system comprises three ML models, one agent LLM, and one RAG layer:

Component	Technology / Model	Type	Purpose
Model 1 — ChestVision	ViT-B/16 fine-tuned via timm (selected); compared with ResNet50V2 & EfficientNetV2B0	Transfer Learning (timm pretrained)	Multi-label X-ray classification (14 conditions); ViT-B/16 selected as final model → ONNX
Model 2 — ClinicalFusion	MLP tabular branch + frozen BioClinical-ModernBERT-large	Hybrid Model	Diagnosis (5 categories) + Severity (0–3) → ONNX
Model 3 — OpenMed NER	OpenMed-NER-DiseaseDetect-SuperClinical-434M	Pretrained + Fine-tuned	Biomedical NER on clinical notes → ONNX
Agent LLM	google/medgemma-27b-it	Instruction-tuned LLM	Final diagnosis reasoning via HuggingFace Inference API
RAG Layer	A-Z Medical Encyclopedia + ChromaDB	Post-decision QA	Supporting context enrichment after diagnosis
Application	Next.js 14 + LangGraph.js + Clerk + NeonDB	Full-stack app	Patient-facing UI, agent orchestration, appointments

3. Phase-by-Phase Timeline

Phase 1: Data Preparation | 20 May – 2 Jun 2026 | Weeks 1 – 2

Objective: Establish clean, split data pipelines for all three models. No model training starts until data is verified and processed.

Task	Dates	Status	Deliverable
NIH ChestX-ray14 Preprocessing	20–26 May	Complete	train/val/test CSVs (85,108 images)
Synthea EHR Feature Engineering	20–26 May	Complete	Scaled feature matrix + label encoder
BIO-tag NER Dataset (Synthea Notes)	27 May–2 Jun	Pending	BIO-tagged NER dataset for Model 3 fine-tuning
Medical Encyclopedia → ChromaDB Setup	3–9 Jun	Pending	Populated ChromaDB collection (RAG)

Phase 2: Model 3 — OpenMed NER (Pretrained + Transfer Learning) | 27 May – 16 Jun 2026 | Weeks 2 – 4

Objective: Fine-tune the OpenMed-NER-DiseaseDetect-SuperClinical-434M pretrained model on Synthea chest/cardiovascular notes. This model must be completed first as its encoder feeds directly into Model 2 training.

Task	Dates	Status	Deliverable
OpenMed NER Fine-tuning	27 May–9 Jun	Pending	Fine-tuned NER weights on Synthea notes
NER Evaluation (seqeval F1)	10–12 Jun	Pending	Entity F1 ≥ 0.82 ; transfer gain ≥ 3 F1 pts
ONNX Export — NER Model (INT8)	13–16 Jun	Pending	Quantised ONNX model ~150MB

Phase 3: Model 2 — ClinicalFusion (Hybrid Model) | 10 Jun – 1 Jul 2026 | Weeks 3 – 6

Objective: Build the hybrid MLP tabular branch + frozen BioClinical-ModernBERT-large model trained on Synthea EHR data. Produces diagnosis classification (5 categories) and severity scoring (0–3).

Task	Dates	Status	Deliverable
XGBoost Tabular Baseline Training	10–17 Jun	Pending	XGBoost baseline; GridSearchCV results
Fusion Network + BioClinical-ModernBERT-large Integration	17 Jun–1 Jul	Pending	Trained fusion model; ablation table
ClinicalFusion Evaluation & Ablation	24–28 Jun	Pending	Macro F1 ≥ 0.80 ; false-low rate $< 2\%$

Task	Dates	Status	Deliverable
ONNX Export — ClinicalFusion	29 Jun–1 Jul	Pending	Validated ONNX pipeline ~120MB

Phase 4: Model 1 — ChestVision (Transfer Learning (timm pretrained)) | 17 Jun – 15 Jul 2026 | Weeks 4 – 8

Objective: Train a custom ResNet-50 (and optionally ViT-B/16) from scratch on NIH ChestX-ray14 for multi-label thoracic condition classification. GPU-intensive — scheduled to run overnight in parallel with application build.

Task	Dates	Status	Deliverable
Fine-tune ResNet50V2, ViT-B/16, EfficientNetV2B0 (two-phase timm)	17 Jun–8 Jul	Pending	Best checkpoint; training curves; Optuna report
Grad-CAM & Explainability	8–10 Jul	Pending	Heatmap overlays; localisation AUC
ChestVision Evaluation (AUC-ROC)	8–11 Jul	Pending	Mean AUC ≥ 0.82 ; per-class AUC ≥ 0.75
ONNX Export + INT8 Quantisation (All Models)	11–15 Jul	Pending	All 3 models as validated ONNX files < 250MB

Phase 5: Application Build | 17 Jun – 19 Jul 2026 | Weeks 5 – 9

Objective: Build the full-stack Next.js 14 application with Clerk authentication, NeonDB, LangGraph.js agent, and MedGemma-27B-IT integration. Runs in parallel with model training phases.

Task	Dates	Status	Deliverable
Next.js 14 + Clerk + NeonDB Scaffold	17 Jun–1 Jul	Pending	Auth-gated app with DB schema and routes
ONNX Inference API Routes (×3)	1–8 Jul	Pending	3 working model inference endpoints
LangGraph Agent + MedGemma-27B-IT	8–15 Jul	Pending	11-node stateful diagnostic agent
ChromaDB RAG Integration	8–12 Jul	Pending	Top-3 encyclopedia retrieval wired to agent
Google Calendar + Gmail API	12–19 Jul	Pending	Appointment booking and email confirmation flow

Phase 6: Production Deployment | 15 – 22 Jul 2026 | Weeks 9 – 10

Objective: End-to-end testing and production deployment to Vercel. CI/CD pipeline via GitHub Actions. All ONNX models validated within Vercel 250MB function size limits.

Task	Dates	Status	Deliverable
E2E Testing & QA (Playwright / Jest)	15–19 Jul	Pending	Smoke test with synthetic patient scenario
Vercel Deployment + CI/CD Pipeline	19–22 Jul	Pending	Live production URL; automated ONNX validation

Phase 7: Report Writing & Submission | 15 – 24 Jul 2026 | Weeks 9 – 10

Objective: Final report writing, presentation preparation, and project submission. Runs in parallel with production deployment.

Task	Dates	Status	Deliverable
Report Writing & Presentation Prep	15–24 Jul	Pending	Final report + presentation slides submitted by 24 Jul

4. Key Milestones

Milestone	Target Date	Project Week
Phase 1 Complete	2 Jun 2026	End of Week 2
OpenMed NER ONNX Export Ready	16 Jun 2026	End of Week 4
ClinicalFusion Model Validated	1 Jul 2026	End of Week 6
All 3 ONNX Models Quantised & Ready	15 Jul 2026	End of Week 8
Full Agent End-to-End Test Passing	19 Jul 2026	Week 9
Production Deployment Live (Vercel)	22 Jul 2026	Week 10
Final Report & Project Submission	24 Jul 2026	Week 10

5. Project Gantt Chart

The Gantt chart below illustrates the full 10-week schedule, showing task dependencies, parallel workstreams, and the production deployment phase immediately before the submission deadline.

